

PROJECT TITLE: OptiCrop: Smart Agricultural Production Optimization Engine

1. PROJECT WORK BREAKDOWN TIMELINE (GANTT TRACKING MATRIX):

The development process of the OptiCrop Production Engine was planned over a structured linear engineering lifecycle:

*** Milestone 1: Requirement Gathering & Scope Identification (Week 1)**

- Core Activity: Finalized parameter scope (N, P, K, Temp, Humidity, pH, Rainfall). Built problem-solution architecture matrices.

*** Milestone 2: Data Engineering & Model Optimization (Week 2)**

- Core Activity: Gathered historical environmental mapping dataset. Developed, trained, and validated the Sklearn Random Forest Classifier binary engine. Saved standalone crop_model.pkl file.

*** Milestone 3: Interface Prototyping & Backend Service Setup (Week 3)**

- Core Activity: Built Flask web application framework layout (app.py). Developed interactive frontend inputs validation views (index.html) using Bootstrap v5 structures.

*** Milestone 4: Operational System Integration & Functional Tests (Week 4)**

- Core Activity: Connected real-time frontend user form inputs directly to the underlying saved machine learning classifier array pipeline. Conducted functional accuracy checks and verification scenarios. Created documentation artifacts.